ONEDRAFT

CAD-AI — Aria Powered

OneDraft End-to-End System Integration Specification

Version 1.0

Document: OneDraft End-to-End System Integration Specification
System: OneDraft CAD-AI
Intelligence Layer: Aria
Document Class: Architectural & Implementation Stack Specification
Status: Master Architectural Specification
Predecessor: Product Configuration & Administrative Control Plane Architecture Specification
Successor: OneDraft End-to-End Implementation, Verification & Operational Lifecycle

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1. PURPOSE

The OneDraft End-to-End System Integration Specification establishes the complete system integration architecture for OneDraft CAD-AI.

This specification consolidates the preceding architectural stacks into one coherent platform architecture and defines how foundational services, domain services, computational engines, compliance systems, documentation systems, runtime infrastructure, security, AI, user interfaces, commercial systems, administrative controls and external integrations operate together.

The purpose of this document is to establish the complete architectural relationship between all OneDraft subsystems and demonstrate the progression from project intake through computational modeling, regulatory verification, documentation generation, review, acceptance, artifact delivery and operational lifecycle.

2. SYSTEM IDENTITY

System: OneDraft CAD-AI

Intelligence Layer: Aria

Primary Function: AI-assisted architectural project specification, computational modeling, regulatory analysis, validation, coordination, documentation generation, review and acceptance.

Architectural Principle: Aria assists and orchestrates; deterministic domain, geometry, compliance, validation and documentation systems establish authoritative computational results.

Primary Output: A validated architectural document package generated from structured project requirements and governed through controlled project lifecycle states.

3. COMPLETE ARCHITECTURAL STACK

The OneDraft architecture consists of the following integrated stacks:

```
1.0  FOUNDATIONAL TECHNICAL SPECIFICATION
2.0  SYSTEM ARCHITECTURE SPECIFICATION
3.0  MVP TECHNICAL STACK & REPOSITORY ARCHITECTURE
4.0  API & DOMAIN SCHEMA SPECIFICATION
5.0  DATABASE SCHEMA & OPENAPI CONTRACT
6.0  EXECUTABLE FOUNDATION IMPLEMENTATION SPECIFICATION
7.0  EXECUTABLE APPLICATION STACK & RUNTIME ARCHITECTURE
8.0  EXECUTABLE SOFTWARE STACK & FOUNDATION IMPLEMENTATION
9.0  EXECUTABLE APPLICATION STACK & IMPLEMENTATION ARCHITECTURE
10.0 EXECUTABLE APPLICATION STACK & IMPLEMENTATION ARCHITECTURE
11.0 EXECUTABLE APPLICATION & DOMAIN SERVICE LAYER
12.0 EXECUTABLE REPOSITORY FOUNDATION & CORE APPLICATION RUNTIME
13.0 APPLICATION RUNTIME, SERVICE ARCHITECTURE & EXECUTION STACK
14.0 COMPUTATIONAL PROJECT MODEL & GEOMETRY ENGINE
15.0 EXECUTABLE APPLICATION STACK & SERVICE ARCHITECTURE
16.0 RUNTIME & APPLICATION ORCHESTRATION STACK
17.0 GEOMETRY & PARAMETRIC MODEL ENGINE
18.0 COMPLIANCE & REGULATORY ENGINE
19.0 VALIDATION & VERIFICATION ENGINE
20.0 VALIDATION & COORDINATION ENGINE
21.0 DOCUMENTATION & DRAWING GENERATION ENGINE
22.0 PROJECT REVIEW, REDLINE, ACCEPTANCE & APPROVAL ENGINE
23.0 RUNTIME ORCHESTRATOR
24.0 JOB & WORKER EXECUTION SYSTEM
25.0 ARTIFACT & STORAGE MANAGEMENT ENGINE
26.0 INTEGRATION & EVENT BUS
27.0 IDENTITY, ACCESS & SECURITY SYSTEM
28.0 IDENTITY, SECURITY & AUTHORIZATION SYSTEM
29.0 IDENTITY, ACCESS CONTROL & SECURITY SYSTEM
30.0 OBSERVABILITY, TELEMETRY & SYSTEM HEALTH
31.0 TESTING, QUALITY ASSURANCE & VERIFICATION SYSTEM
32.0 CI/CD, BUILD & RELEASE ENGINEERING
33.0 INFRASTRUCTURE & DEPLOYMENT ARCHITECTURE
34.0 CONFIGURATION, SECRETS & ENVIRONMENT MANAGEMENT
35.0 BACKUP, DISASTER RECOVERY & BUSINESS CONTINUITY
36.0 DATA LIFECYCLE, RETENTION & PROVENANCE SYSTEM
37.0 AI INTELLIGENCE & MODEL PROVIDER ARCHITECTURE
38.0 AI EVALUATION, GUARDRAILS & DETERMINISTIC COMMAND VERIFICATION
39.0 STANDARDS, ARCHITECTURAL GRAPHICS & CAD INTEROPERABILITY
40.0 IMPORT, EXPORT & EXTERNAL DATA EXCHANGE
41.0 FRONTEND & USER INTERACTION ARCHITECTURE
42.0 PROJECT LIFECYCLE & STATE MACHINE
43.0 BILLING, SUBSCRIPTION & ENTITLEMENT ARCHITECTURE
44.0 MULTI-TENANT ORGANIZATION & PROJECT ISOLATION
45.0 API GATEWAY, RATE LIMITING & SERVICE PROTECTION
46.0 PERFORMANCE, SCALABILITY & CAPACITY ARCHITECTURE
47.0 DISASTER TESTING, SECURITY TESTING & OPERATIONAL READINESS
48.0 DEVELOPER PLATFORM, SDK & EXTENSION ARCHITECTURE
49.0 PRODUCT CONFIGURATION & ADMINISTRATIVE CONTROL PLANE
50.0 END-TO-END SYSTEM INTEGRATION
```

Each stack establishes a defined architectural responsibility while remaining integrated with the contracts, interfaces and invariants established throughout the platform.

4. SYSTEM OF SYSTEMS

OneDraft shall operate as a system of coordinated subsystems rather than as an unrestricted monolithic application.

The principal architecture is:

```
                    ONEDRAFT
                       │
        ┌──────────────┼──────────────┐
        │              │              │
     FRONTEND        API           ARIA
        │              │              │
        └──────────────┼──────────────┘
                       │
                 APPLICATION
                    SERVICES
                       │
        ┌──────────────┼──────────────┐
        │              │              │
      DOMAIN        RUNTIME       SECURITY
        │              │              │
        ├──────────────┼──────────────┤
        │              │              │
    GEOMETRY      JOB SYSTEM      EVENTS
        │              │              │
    COMPLIANCE      WORKERS      ARTIFACTS
        │              │              │
   VALIDATION      STORAGE       DATABASE
        │              │              │
 DOCUMENTATION   INFRASTRUCTURE  OBSERVABILITY
```

5. AUTHORITATIVE SYSTEM LAYERS

OneDraft shall distinguish between systems that propose, systems that compute and systems that establish authoritative state.

Aria proposes, interprets, reasons and orchestrates.

Application Services coordinate domain operations.

Domain Services enforce business rules.

Geometry Engine computes architectural geometry.

Compliance Engine determines applicable regulatory requirements.

Validation Engine evaluates deterministic project conditions.

Documentation Engine generates architectural documentation.

Database persists authoritative transactional state.

Artifact Storage preserves generated files and immutable project outputs.

This separation prevents AI-generated suggestions from becoming authoritative merely because they were produced by the intelligence layer.

6. END-TO-END PROJECT FLOW

The primary OneDraft project lifecycle shall follow:

```
PROJECT INTAKE
      ↓
REQUIREMENTS
      ↓
JURISDICTION IDENTIFICATION
      ↓
CODE / REGULATORY ANALYSIS
      ↓
PROJECT SPECIFICATION
      ↓
COMPUTATIONAL PROJECT MODEL
      ↓
PARAMETRIC GEOMETRY
      ↓
COMPLIANCE COMPUTATION
      ↓
VALIDATION
      ↓
COORDINATION
      ↓
DOCUMENT GENERATION
      ↓
AI / SYSTEM REVIEW
      ↓
CUSTOMER REVIEW
      ↓
REDLINES
      ↓
REVISION
      ↓
REVALIDATION
      ↓
FINAL DOCUMENT PACKAGE
      ↓
ACCEPTANCE
      ↓
ARTIFACT RELEASE
```

No stage shall silently bypass a required predecessor.

7. PROJECT INTAKE

The project begins through structured intake.

Inputs may include:

Project Description

Site Information

Jurisdiction

Building Program

Occupancy

Area Requirements

Levels

Structural Requirements

Mechanical Requirements

Electrical Requirements

Accessibility Requirements

Client Requirements

Existing Documentation

Survey Data

Reference Files

The intake system converts unstructured and structured inputs into a controlled project specification.

8. ARIA REQUIREMENT INTERPRETATION

Aria shall assist in transforming project information into structured requirements.

Aria may:

Identify Missing Information

Normalize Requirements

Detect Contradictions

Request Clarification

Classify Project Requirements

Construct Project Context

Propose Project Parameters

Aria output shall remain subject to deterministic validation.

9. JURISDICTION ANALYSIS

The Compliance & Regulatory Engine shall identify applicable jurisdictional requirements.

The process shall consider:

Geographic Location

Authority Having Jurisdiction

Building Classification

Occupancy

Construction Type

Building Area

Building Height

Number of Storeys

Applicable Code Editions

Referenced Standards

The resulting compliance context becomes part of the project record.

10. REGULATORY PROVENANCE

Regulatory decisions shall maintain provenance.

The system shall preserve relationships between:

```
JURISDICTION
   ↓
AUTHORITY
   ↓
CODE SOURCE
   ↓
CODE VERSION
   ↓
RULE
   ↓
APPLICABILITY
   ↓
PROJECT RESULT
```

Regulatory analysis shall therefore remain traceable to its source and effective version.

11. COMPUTATIONAL PROJECT MODEL

The Project Model shall provide the canonical computational representation of the architectural project.

The model shall represent:

Site

Building

Levels

Spaces

Assemblies

Elements

Openings

Circulation

Systems

Relationships

Constraints

Dimensions

Materials

Attributes

The model shall support versioning and deterministic regeneration.

12. GEOMETRY GENERATION

The Geometry & Parametric Model Engine shall transform project parameters into computational geometry.

Geometry generation shall support:

Parametric Constraints

Spatial Relationships

Dependencies

Dimensions

Assemblies

Openings

Levels

Sections

Views

Changes shall propagate through the dependency graph where applicable.

13. COMPLIANCE COMPUTATION

The Compliance Engine shall evaluate project conditions against applicable regulatory rules.

Representative checks include:

Area

Height

Occupancy

Egress

Travel Distance

Exits

Fire Separation

Accessibility

Spatial Requirements

Required Clearances

Results shall be represented as structured verification records.

14. VALIDATION

The Validation & Verification Engine shall execute deterministic validation.

Validation shall classify results according to defined severity.

Representative classifications include:

PASS

WARNING

ERROR

CRITICAL

Validation shall operate against authoritative project and regulatory state.

15. COORDINATION

The Coordination Engine shall verify consistency across project representations.

Coordination may include:

Model-to-Drawing

Geometry-to-Documentation

Level Coordination

Dimension Coordination

Space Coordination

System Coordination

Schedule Coordination

Conflicts shall become explicit validation results rather than remaining implicit within generated documents.

16. AI COMMAND ARCHITECTURE

Aria commands shall follow:

```
USER INTENT
      ↓
ARIA INTERPRETATION
      ↓
STRUCTURED COMMAND
      ↓
COMMAND VALIDATION
      ↓
AUTHORIZATION
      ↓
DOMAIN SERVICE
      ↓
COMPUTATION
      ↓
VALIDATION
      ↓
RESULT
```

Aria shall not directly mutate authoritative project state outside this controlled command path.

17. DOCUMENT GENERATION

The Documentation & Drawing Generation Engine shall consume validated project state.

Outputs may include:

Plans

Elevations

Sections

Details

Schedules

Annotations

Dimensions

Title Blocks

Drawing Sheets

Document Packages

Documentation generation shall reference the authoritative Project Model.

18. CAD & BIM INTEROPERABILITY

OneDraft shall maintain controlled interoperability with:

DXF

DWG

IFC

BIM Data

PDF

Images

Survey Data

Structured Specifications

Import and export operations shall preserve provenance and identify transformations where applicable.

19. REVIEW & REDLINES

Generated documentation shall enter the Review, Redline, Acceptance & Approval workflow.

Users may:

Review

Comment

Redline

Request Revision

Accept

Reject

Approve

Every revision shall remain associated with the applicable project and document version.

20. REVISION CYCLE

The revision loop shall operate as:

```
DOCUMENTATION
      ↓
REVIEW
      ↓
REDLINE
      ↓
REVISION REQUEST
      ↓
ARIA / APPLICATION PROCESSING
      ↓
MODEL UPDATE
      ↓
RECOMPUTATION
      ↓
VALIDATION
      ↓
DOCUMENT REGENERATION
      ↓
REVIEW
```

A revision shall not be considered final until affected computations and documentation have been revalidated.

21. PROJECT STATE MACHINE

The Project Lifecycle architecture shall govern project progression.

Representative states include:

```
INTAKE
  ↓
SPECIFICATION
  ↓
MODELING
  ↓
COMPLIANCE
  ↓
VALIDATION
  ↓
DOCUMENTATION
  ↓
REVIEW
  ↓
REVISION
  ↓
ACCEPTANCE
  ↓
FINALIZED
  ↓
```

Only valid state transitions shall be permitted.

22. RUNTIME ORCHESTRATION

The Runtime Orchestrator coordinates system execution.

It shall manage:

Workflow Sequencing

Service Invocation

Command Routing

Job Creation

State Transitions

Failure Handling

Completion

The orchestrator shall not replace domain services as the authority for domain rules.

23. JOB EXECUTION

Long-running operations shall execute through the Job & Worker Execution System.

Jobs may include:

Geometry Generation

Validation

Compliance Analysis

Drawing Generation

Rendering

Import

Export

AI Processing

Jobs shall have persistent state and controlled retry behavior.

24. EVENT ARCHITECTURE

The Integration & Event Bus shall provide asynchronous coordination.

Events may communicate:

Project Changes

Model Changes

Validation Completion

Drawing Completion

Artifact Creation

Review Events

Billing Events

Security Events

Events shall be versioned and traceable.

25. ARTIFACT MANAGEMENT

Generated artifacts shall be stored through the Artifact & Storage Management Engine.

Each artifact shall maintain:

Artifact Identifier

Project

Version

Type

Format

Hash

Creation Time

Source

Status

Artifacts shall remain associated with their originating project state.

26. DATABASE ARCHITECTURE

PostgreSQL shall provide authoritative transactional persistence.

The database shall preserve:

Organizations

Users

Projects

Models

Revisions

Compliance Results

Validation Results

Drawings

Jobs

Events

Artifacts Metadata

Billing

Configuration

Database transactions shall protect domain consistency.

27. SECURITY ARCHITECTURE

Security shall operate across every system layer.

Security boundaries shall include:

Authentication

Authorization

Tenant Isolation

Project Isolation

API Protection

Credential Management

Artifact Security

Administrative Security

No subsystem shall bypass the security architecture.

28. MULTI-TENANCY

Each organization shall operate within a controlled tenant boundary.

Tenant isolation shall apply to:

Users

Projects

Models

Drawings

Artifacts

Jobs

AI Context

Events

Billing

Configuration

Cross-tenant access shall require explicit authorized administrative mechanisms.

29. API GATEWAY

The API Gateway provides the primary external application boundary.

It shall provide:

Routing

Authentication Integration

Rate Limiting

Request Validation

Service Protection

Correlation

Traffic Management

External API requests shall enter through controlled interfaces.

30. PERFORMANCE & SCALABILITY

The platform shall support horizontal scaling of appropriate stateless services.

Performance management shall include:

Latency

Throughput

Concurrency

Queue Depth

Worker Utilization

Database Capacity

Storage Capacity

AI Capacity

Long-running workloads shall be isolated from interactive workloads where practical.

31. OBSERVABILITY

Observability shall provide system-wide visibility.

The platform shall collect:

Logs

Metrics

Traces

Health Checks

Alerts

Audit Events

Performance Data

Security Events

Correlation identifiers shall permit tracing a transaction across service boundaries.

32. DATA PROVENANCE

Data lineage shall connect source information to generated outputs.

Representative chain:

```
USER INPUT
   ↓
PROJECT REQUIREMENT
   ↓
PROJECT MODEL
```

```
    ↓
GEOMETRY
    ↓
COMPLIANCE
    ↓
VALIDATION
    ↓
DRAWING
    ↓
ARTIFACT
```

The platform shall preserve provenance where required for auditability and reproducibility.

33. DATA LIFECYCLE

Data shall progress through controlled lifecycle states.

The lifecycle shall govern:

Creation

Modification

Versioning

Retention

Archival

Deletion

Recovery

Provenance

Lifecycle rules shall apply according to resource class and applicable policy.

34. BACKUP & RECOVERY

Critical system state shall be protected through backup and recovery architecture.

Recovery shall include:

Database Restoration

Artifact Restoration

Configuration Restoration

Service Recovery

Queue Recovery

Recovery procedures shall be tested rather than assumed.

35. TESTING

Testing shall span the complete architecture.

Test categories shall include:

Unit

Integration

Contract

Domain

Geometry

Compliance

Validation

Documentation

Security

Performance

Disaster Recovery

End-to-End

No individual subsystem shall be considered production-ready solely because its local tests pass.

36. CI/CD

The CI/CD architecture shall provide controlled progression from source code to production.

```
SOURCE
 ↓
BUILD
 ↓
UNIT TEST
 ↓
INTEGRATION TEST
 ↓
CONTRACT TEST
 ↓
SECURITY TEST
 ↓
PERFORMANCE TEST
 ↓
STAGING
 ↓
SYSTEM VALIDATION
 ↓
RELEASE
 ↓
```

MONITORING

Releases shall preserve version traceability across software and configuration.

37. INFRASTRUCTURE

Infrastructure shall provide:

Containers

Application Services

PostgreSQL

Queues

Workers

Object Storage

Networking

Secrets

Observability

Deployment Environments

Infrastructure shall be reproducible and version-controlled where practical.

38. AI ARCHITECTURE

Aria shall operate as the intelligence and orchestration layer.

Aria capabilities include:

Natural Language Interpretation

Requirement Analysis

Context Construction

Project Reasoning

Command Generation

Revision Assistance

Workflow Orchestration

Document Analysis

User Interaction

AI provider abstraction shall permit controlled model selection.

39. AI SAFETY & DETERMINISTIC VERIFICATION

AI outputs shall pass through appropriate controls before affecting authoritative system state.

The governing model is:

```
ARIA OUTPUT
    ↓
STRUCTURAL VALIDATION
    ↓
AUTHORIZATION
    ↓
DOMAIN VALIDATION
    ↓
GEOMETRY / RULE COMPUTATION
    ↓
DETERMINISTIC VERIFICATION
    ↓
AUTHORITATIVE STATE
```

The intelligence layer shall never be treated as a substitute for deterministic architectural computation where deterministic computation is available.

40. CAD GRAPHICS & DOCUMENT STANDARDS

Generated architectural documentation shall use controlled graphic standards.

Standards may govern:

Sheet Size

Layers

Lineweights

Symbols

Text

Dimensions

Annotations

Title Blocks

Drawing Organization

Standards shall be configurable through the Administrative Control Plane.

41. IMPORT & EXPORT

External data shall enter through controlled exchange pipelines.

The pipeline shall provide:

Format Detection

Validation

Normalization

Transformation

Provenance

Error Handling

Artifact Registration

Exports shall identify their source project and generation context.

42. FRONTEND

The frontend shall provide the primary user interaction environment.

Capabilities shall include:

Project Workspace

Project Intake

Model Viewer

Drawing Viewer

Redline Interface

Review Interface

Aria Interface

Artifact Management

Project Administration

The frontend shall consume public application interfaces rather than directly accessing internal databases.

43. BILLING & ENTITLEMENTS

Commercial controls shall govern:

Subscriptions

Plans

Usage

Quotas

Entitlements

Metering

Organization Billing

Billing restrictions shall integrate with authorization and product configuration without corrupting project state.

44. DEVELOPER PLATFORM

The Developer Platform shall expose controlled extension interfaces.

These include:

REST API

Python SDK

TypeScript SDK

Webhooks

Events

Plugins

CLI

External Integrations

Developer access shall remain subject to API, identity, authorization and tenant boundaries.

45. ADMINISTRATIVE CONTROL PLANE

The Administrative Control Plane shall govern:

Feature Flags

Jurisdictions

Code Sources

Code Manifests

Templates

CAD Standards

AI Configuration

System Limits

Security Policies

Product Configuration

Administrative changes shall be versioned, validated and auditable.

46. END-TO-END INTEGRATION CONTRACT

The complete OneDraft platform shall operate according to the following integration contract:

```
USER
  ↓
FRONTEND
  ↓
API GATEWAY
  ↓
IDENTITY / AUTHORIZATION
  ↓
PROJECT / APPLICATION SERVICES
  ↓
RUNTIME ORCHESTRATOR
  ↓
ARIA
  ↓
DOMAIN SERVICES
  ↓
PROJECT MODEL
  ↓
GEOMETRY
  ↓
COMPLIANCE
  ↓
VALIDATION
  ↓
DOCUMENTATION
  ↓
REVIEW / REDLINE
  ↓
REVALIDATION
  ↓
ARTIFACT GENERATION
  ↓
STORAGE
```

```
  ↓
ACCEPTANCE
  ↓
FINAL PACKAGE
```

At every stage, security, provenance, observability and lifecycle controls remain active.

47. SYSTEM-WIDE ARCHITECTURAL INVARIANTS

The following principles apply across the entire OneDraft architecture.

Authoritative State: The Project Model and transactional domain services remain authoritative.

AI Subordination: Aria assists, interprets and orchestrates but does not bypass deterministic validation.

Determinism: Computational results shall be reproducible where deterministic inputs and versions are provided.

Traceability: Major system operations shall be traceable.

Tenant Isolation: Organizations shall remain isolated.

Security: Authentication and authorization shall apply across all resource boundaries.

Versioning: Material project, rule, configuration and artifact changes shall be versioned.

Provenance: Significant outputs shall maintain source lineage.

Validation: Critical operations shall be validated before becoming authoritative.

Recoverability: Critical system state shall be recoverable.

Scalability: The architecture shall support increasing workload without fundamental redesign of domain boundaries.

Extensibility: External integrations shall use controlled contracts.

Operational Visibility: System behavior shall remain observable.

Configuration Governance: Administrative configuration shall remain controlled and auditable.

48. INTEGRATION VERIFICATION

The completed platform shall be verified as an integrated system.

Verification shall include:

Project Intake → Project Model

Project Model → Geometry

Geometry → Compliance

Compliance → Validation

Validation → Documentation

Documentation → Review

Review → Revision

Revision → Revalidation

Finalization → Artifact Package

Artifact Package → Acceptance

API → Application Services

Application Services → Runtime

Runtime → Workers

Workers → Storage

AI → Commands

Commands → Domain Validation

Security → Every Boundary

Observability → Every Service

49. ARCHITECTURAL COMPLETION

With Stack 50.0, the OneDraft architectural register reaches its defined end-to-end integration layer.

The resulting architecture establishes:

```
FOUNDATION
     +
DOMAIN
     +
PROJECT MODEL
     +
GEOMETRY
     +
COMPLIANCE
     +
VALIDATION
     +
DOCUMENTATION
     +
REVIEW
     +
RUNTIME
     +
JOBS
```

```
     +
ARTIFACTS
     +
EVENTS
     +
SECURITY
     +
OBSERVABILITY
     +
TESTING
     +
INFRASTRUCTURE
     +
DATA GOVERNANCE
     +
ARIA
     +
CAD / BIM
     +
DATA EXCHANGE
     +
FRONTEND
     +
LIFECYCLE
     +
BILLING
     +
MULTI-TENANCY
     +
API PROTECTION
     +
SCALABILITY
     +
OPERATIONAL READINESS
     +
DEVELOPER PLATFORM
     +
ADMINISTRATION
     ↓
ONEDRAFT
```

The 50-stack architecture therefore represents one continuous technical progression rather than fifty independent systems.

50. COMPLETION STATUS

The OneDraft End-to-End System Integration Specification establishes the master integration architecture for the complete OneDraft CAD-AI platform.

The specification consolidates the preceding architectural layers into one coherent system encompassing foundational architecture, executable application services, domain modeling, computational geometry, regulatory compliance, deterministic validation, architectural documentation, project review, runtime orchestration, asynchronous execution, artifact management, event integration, identity and authorization, observability, testing, CI/CD, infrastructure, data governance, Aria intelligence, CAD/BIM interoperability, external data exchange, frontend interaction, project lifecycle, billing, multi-tenancy, API protection, scalability, disaster recovery, operational readiness, developer extensibility and administrative control.

The resulting platform architecture establishes the complete progression:

```
SPECIFICATION
      ↓
IMPLEMENTATION
      ↓
COMPUTATION
      ↓
VALIDATION
      ↓
DOCUMENTATION
      ↓
REVIEW
      ↓
ACCEPTANCE
      ↓
OPERATION
      ↓
CONTINUOUS GOVERNANCE
```

OneDraft is therefore architecturally defined as an integrated CAD-AI platform in which Aria provides the intelligence and orchestration layer while deterministic computational, regulatory, validation, documentation, security and lifecycle systems establish authoritative system behavior.

Status: END-TO-END ARCHITECTURALLY DEFINED

Architecture Register: 1.0–50.0 COMPLETE

System: ONEDRAFT CAD-AI

Intelligence Layer: ARIA

Architectural State: COMPLETE INTEGRATED SYSTEM ARCHITECTURE